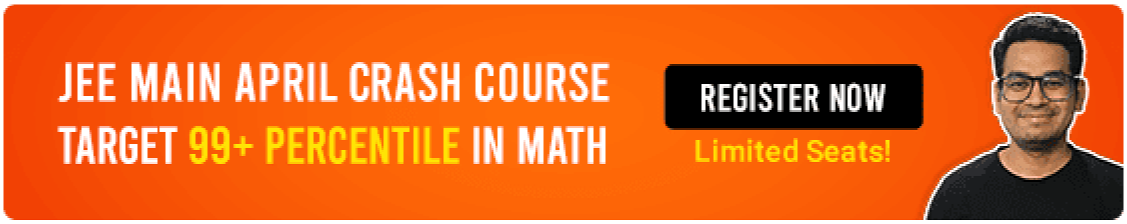
Master Math for JEE Main & JEE Advanced

Crash Courses designed specifically for students who want to improve their percentile & score in upcoming JEE Main & JEE Advanced exam. **Tap on the banners to know more.**

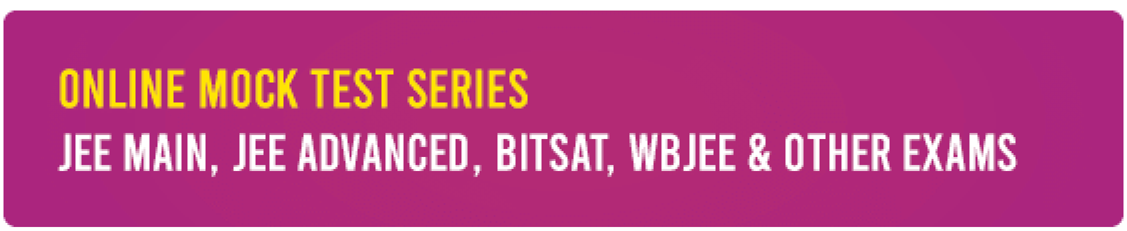
For JEE Main 2020 April



For JEE Advanced 2020

Online Mock Test Series

Take online mock tests for a solid practice for your upcoming JEE Main, JEE Advanced, BITSAT & other competitive exams. All the subjects are included. **Tap on the banner.**



Free Study Materials

Download PDF of study materials like notes, formula sheets, practice questions, previous year question papers from our website. Visit www.mathongo.com now!

JEE Mains 2020 Jan Chapter wise Question Bank

EMI

Q1

A long solenoid of radius R carries a time dependent current $I = I_0 t(1 - t)$. A ring of radius 2R is placed coaxially near its centre. During the time interval $0 \le t \le 1$, the induced current I_R and the induced emf V_R in the ring vary as:

(1) current will change its direction and its emf will be zero at t = 0.25sec.
(2) current will not change its direction & emf will be maximum at t = 0.5sec
(3) current will not change direction and emf will be zero at 0.25sec.
(4) current will change its direction and its emf will be zero at t = 0.5sec.

7^{th} Jan Morning

Sol

(4)

$I = I_0 t - I_0 t^2$

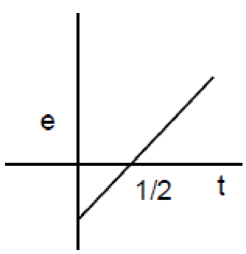
$\phi = BA$

$\phi = \mu_0 n I A$

$V_R = -\dfrac{d\phi}{dt} = -\mu_0 n A I_0 (1 - 2t)$

$V_R = 0$ at $t = \dfrac{1}{2}$

and तथा $I_R = \dfrac{V_R}{\text{Resistance of loop}}$ $\dfrac{V_R}{\text{लुप का प्रतिरोध}}$



Q2

A particle of mass m and positive charge q is projected with a speed of v_0 in y–direction in the presence of electric and magnetic field are in x–direction. Find the instant of time at which the speed of particle becomes double the initial speed.

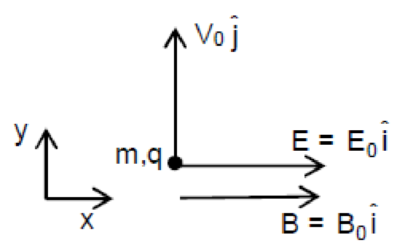
x-दिशा में उपस्थित विद्युत तथा चुम्बकीय क्षेत्र में एक m द्रव्यमान तथा धनात्मक आवेश q का कण v_0 चाल से y–दिशा में प्रक्षेपित किया जाता है। वह समय ज्ञात कीजिए जब कण की चाल प्रारम्भिक चाल की दोगुना हो जाऐगी

(1) $t = \dfrac{mv_0\sqrt{3}}{qE}$ (2) $t = \dfrac{mv_0\sqrt{2}}{qE}$ (3) $t = \dfrac{mv_0}{qE}$ (4) $t = \dfrac{mv_0}{2qE}$

7^{th} Jan Evening

Sol

(1)



As $\vec{v} = v_0\hat{j}$ (magnitude of velocity does not change in y–z plane)

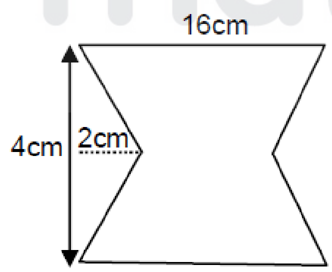
$(2v_0)^2 = v_0^2 + v_x^2$; $v_x = \sqrt{3}v_0$

$\therefore \quad \sqrt{3}v_0 = 0 + \frac{qE}{m}t$; $t = \frac{mv_0\sqrt{3}}{qE}$

Q3

The given loop is kept in a uniform magnetic field perpendicular to plane of loop. The field changes from 1000G to 500G in 5seconds. The average induced emf in loop is–

चुम्बकीय क्षेत्र के लम्बवत् दिया गया लूप रखा गया है यदि क्षेत्र 1000G से 500G तक 5 सेकण्ड में परिवर्तित होता है तो औसत प्रेरित वि.वा.ब ज्ञात करो।



(1) 56 μV (2) 28 μV (3) 30 μV (4) 48 μV

8^{th} Jan Morning

Sol

(1)

$$\varepsilon = \left|-\frac{d\varphi}{dt}\right| = \left|-\frac{AdB}{dt}\right|$$

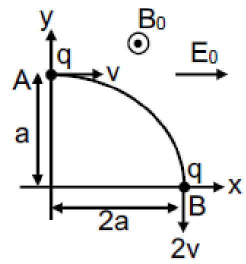
$$= .(16 \times 4 - 4 \times 2)\frac{(1000-500)}{5} \times 10^{-4} \times 10^{-4}$$

$$= 56 \times \frac{500}{5} \times 10^{-8} = 56 \times 10^{-6}\ \text{V}$$

Q4

Which of the following statements are correct for moving charge as shown in figure.

दिये गये चित्र में गतिशील आवेश के लिये नीचे दिया गया कौनसा कथन सत्य है–



(A) Magnitude of electric field $E_0 = \frac{3}{4}\left(\frac{mv^2}{qa}\right)$

(B) Rate of work done at point A is $\frac{3}{4}\frac{mv^3}{a}$

(C) Rate of work done by both fields at point B is zero.

(D) Change in angular momentum w.r.t. origin is 2maV

9th Jan Morning

Sol

(1)

(A) by work energy theorem कार्य ऊर्जा प्रेमय से

$$W_{mag} + W_{ele} = \frac{1}{2}m(2v)^2 - \frac{1}{2}m(v)^2$$

$$0 + qE_0 2a = \frac{3}{2}mv^2$$

$$E_0 = \frac{3}{4}\frac{mv^2}{qa}$$